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Prevalence, incidence, mortality, and years of life lost to diabetes from 1996 to 2024 in Denmark

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ABSTRACT

Introduction Population-based surveillance is essential for quantifying the burden of diabetes and monitoring temporal changes. In Denmark, nationwide health registers enable long-term surveillance of type 1 diabetes (T1D) and type 2 diabetes (T2D). We aimed to estimate trends in prevalence, incidence, mortality and years of life lost to diabetes in Denmark from 1996 to 2024.

Research design and methods We conducted a nationwide register-based cohort study from 1996 to 2024. We used an updated national diabetes register integrating hospital, prescription, clinical, and laboratory data to identify diabetes type and date of onset. Prevalence was estimated annually using a binomial model. Incidence, mortality and standardized mortality ratios (SMR) were analyzed using age-period-cohort models. Analyses were stratified by sex and diabetes type. A multistate model was used to assess the lifetime risk and years of life lost to diabetes.

Results As of January 1, 2025, a total of 366 174 individuals (6.1% of the population) had diabetes identified through national healthcare registers; of these, 8.2% had T1D and 91.8% had T2D. Since 1996, diabetes prevalence more than tripled, largely driven by an overall annual increase in T2D of ~4.1%. T1D prevalence remained stable, with increases in individuals aged <40 years and decreases in older age groups. From 1996 to 2024, a total of 567 510 incident diabetes cases were recorded, of which 4.6% were T1D. T1D incidence declined on average by 1.1% annually, with increasing rates at younger ages and declining rates at older ages. T2D incidence showed a non-linear pattern, with a decline from 2010 to 2015 followed by an increase in recent years, resulting in an overall upward trend from 1996 to 2024. Men had higher T2D incidence than women across age and calendar years. Mortality declined for T1D after 2008 and for T2D until 2012, then stabilized. SMRs decreased for T1D after 2008 but increased for T2D, converging by 2025.

Conclusions In Denmark, diabetes prevalence and incidence increased markedly since 1996, largely driven by T2D. Although mortality declined, particularly for T1D, excess mortality remains, emphasizing the need for continued surveillance and prevention efforts.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ The prevalence of type 1 and type 2 diabetes has increased in many countries in recent decades, while incidence trends have varied, with stabilization or declines reported in some high-income countries in recent years.

⇒ Individuals with diabetes continue to experience substantially higher mortality than the general population.

WHAT THIS STUDY ADDS

⇒ The prevalence and incidence of type 1 diabetes increased among younger individuals (<40 years) but decreased in older age groups, whereas type 2 diabetes prevalence and incidence increased across all age groups.

⇒ Mortality rates decreased substantially for type 1 diabetes after 2008 and decreased for type 2 diabetes until around 2012, after which they stabilized.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Despite declining mortality, the increasing prevalence and incidence of diabetes, primarily driven by type 2 diabetes, with increases in type 1 diabetes among younger age groups, indicate a growing population burden. Continued efforts to prevent diabetes and reduce diabetes-related complications are needed to narrow the mortality gap between people with and without diabetes.

INTRODUCTION

Population-based surveillance is essential for quantifying the burden of diabetes and monitoring temporal trends.[1–3] However, few countries have sufficiently comprehensive data to enable long-term nationwide surveillance, particularly during periods of change. Estimates from high-income countries, covering data up to the mid-2010s, have reported stabilization or declines in diabetes incidence in the later years of the observation period.[4] However, trends appear heterogeneous across settings and time periods, and more recent patterns after the mid-2010s remain uncertain.

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Systematic diabetes monitoring in Denmark began in 2006 with the first national diabetes register.[5] In 2018, the Steno Diabetes Register (DMreg) enabled more accurate differentiation between type 1 diabetes (T1D) and type 2 diabetes (T2D).[6] Since then, nationwide laboratory data have become available, prompting a revised algorithm and updated analyses.

Previous analyses showed that diabetes prevalence in Denmark more than doubled between 1996 and 2016, reflecting increased incidence and lower mortality.[6] Despite improved survival, excess mortality remained substantial.[6 7] Since 2016, several initiatives have transformed diabetes care in Denmark, including the establishment of five Danish Steno Diabetes Centers.[8] In addition, pharmacological treatment and technology have improved. Among individuals with T1D, the uptake of advanced insulin pump therapy, including hybrid closed-loop systems and continuous glucose monitoring systems, has increased.[9–11] For T2D, the introduction of glucagon-like peptide 1 (GLP-1) receptor agonists and sodium-glucose cotransporter 2 (SGLT-2) inhibitors has improved management and outcomes. Trial data have also demonstrated cardiovascular and kidney protection, and guidelines now recommend these drugs for people with diabetes and complications.[12 13] These changes may have influenced recent trends in incidence and mortality.

Monitoring diabetes remains essential for evaluating the disease burden and temporal trends. Therefore, we provide updated nationwide estimates of prevalence, incidence, mortality, and years of life lost (YLL) among individuals with T1D and T2D identified through national healthcare registers between 1996 and 2024.

METHODS

This is a nationwide, register-based cohort study comprising the entire Danish population from 1996 to 2024. All registers used are linked through the Central Person Register number[14] and analyzed on secure servers at Statistics Denmark.

Population data

Population data were obtained from the 'LifeLines' register at Statistics Denmark, comprising all individuals with a registered residence in Denmark between 1986 and 2025. The register includes information on date of birth, migration, and date of death. Because of migration, some persons have multiple records, representing disjoint periods of residence in Denmark.

Diabetes data

We used an updated national diabetes register, DMreg, integrating data from existing national diabetes and patient registers as well as hospital, prescription, clinical, and laboratory data. DMreg included information on diabetes type and diagnosis date for all individuals with identified diabetes from January 1, 1996 to January 1, 2025.

Diabetes classification and date of diagnosis

The inclusion date was determined from the different registers as: (1) the second of the two earliest diabetes diagnosis dates in the National Patient Register (NPR)[15]; or (2) the second of the two earliest purchase dates of insulin or non-insulin antidiabetic medicine in the Danish National Prescription Registry (DNPR),[16] excluding GLP-1 receptor agonist and SGLT-2 inhibitor drugs, because of obesity and heart failure indications, respectively, or one recorded date in each register of NPR and DNPR; or (3) the earliest recorded date of diagnosis in the Danish Adult Diabetes Database (DADD) (2005–2020) or Danish Diabetes Database (2020–2025)[17]; or (4) the earliest date of billing of podiatry for patients with diabetes in the National Health Services Register (NHSR)[18]; or (5) the latest date of two consecutive glycated hemoglobin (HbA1c) ≥48 mmol/mol in the Register of Laboratory Results for Research.[19]

Because the registers have different coverage periods, inclusion dates before January 1, 1996 may not reflect the true onset of diabetes. Since the DNPR, the newest of the main registers (DNPR, NPR and NHSR), was established on January 1, 1995, we used January 1, 1996 as the cut point for DMreg, allowing 1 year of coverage before considering recorded purchases as incident purchases. Hence, only inclusion dates after January 1, 1996 were regarded as valid dates of diabetes diagnosis; earlier cases were regarded as prevalent cases as of January 1, 1996.

Pregnancy periods were identified using the Danish Medical Birth Register, and any diabetes criteria met during pregnancy were excluded to minimize misclassification of gestational diabetes. Metformin dispensations alone before age 40 were disregarded in women with a recorded polycystic ovary syndrome (PCOS) diagnosis, while those after age 40 counted toward diabetes inclusion. In women without a PCOS diagnosis, dispensations between 20 and 40 years were disregarded if metformin was the only glucose-lowering medication.

Diabetes type

Diabetes was classified using the most recent recording of type in the registers. Individuals were classified as T1D if the latest DADD record indicated T1D; otherwise, persons were classified as T2D or other types of diabetes. If DADD data were unavailable, the most recent diagnosis code in NPR was used. Individuals with insulin purchase before age 30 were classified as T1D unless DADD indicated T2D or other types of diabetes. Individuals classified as T1D without insulin purchase within 1 year after inclusion were reclassified as T2D, regardless of recordings in DADD and NPR. All other persons were classified as T2D. Other types of diabetes were analyzed as T2D in the analyses.

A detailed description of the algorithm is provided in the online supplemental material, and the full description is provided elsewhere.[20]

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Statistics

Prevalence

Prevalence was estimated for each January 1 from 1996 through 2025 classifying persons alive on January 1 each year as no diabetes, T1D, or T2D. Presence of T1D and T2D was analyzed by a binomial model with a log link with natural splines to flexibly model the effect of age and calendar time. We graphed the obtained age-specific prevalences for each sex and type of diabetes. We also fitted models with a linear effect of date of prevalence to provide an overall trend in prevalence, as well as in interaction with a spline effect of age to produce calendar time trends as a function of age for each combination of diabetes type and sex.

Follow-up intervals

Each person was followed from the earliest of January 1, 1996, date of birth, or date of immigration, until the earliest of January 1, 2025, date of death, or date of emigration. Follow-up time was recorded for the states no diabetes, T1D, or T2D, with transitions from no diabetes to T1D or T2D recorded as the date of diagnosis. Persons could contribute with multiple non-overlapping follow-up periods if they emigrated and subsequently reimmigrated. All analyses were done separately for men and women, and for T1D, T2D and no diabetes.

Incidence

Incidence rates of T1D and T2D were analyzed using follow-up time from entry into the population in the state of no diabetes until the date of diabetes diagnosis, divided into 1-year intervals. Rates were modeled using Poisson regression with natural spline effects of current age, calendar time, and date of birth, corresponding to an age-period-cohort framework.[21]

Mortality

Analysis of mortality rates was based on follow-up in either of the states no diabetes, T1D or T2D, until the date of death or end of study (December 31, 2024). The analyses of mortality were restricted to a population without diabetes as of January 1, 1996. Standardized mortality ratios (SMRs) were analyzed in the same way as mortality rates using the expected number of deaths derived from the mortality model for individuals without diabetes. Mortality rate ratio comparing T1D and T2D was also calculated.

Predicted incidence rates, mortality rates, and SMRs were presented as functions of age at selected calendar times and calendar times at selected ages. If we had modeled rates only using effects of age and calendar time, we would have seen collections of parallel curves, but because we allow interaction via date of birth, the resulting curves allow different age shapes at different dates and vice versa. Average annual changes in each rate were obtained from age-period models[18] for the full follow-up period (1996–2024), and separately for the period 2010–2024 in a sensitivity analysis to assess the trends in the latest period.

Years of life lost

YLLs due to T1D were estimated as the difference in expected residual lifetime (ERL) between a person with T1D and a person without diabetes of the same sex and age, and similarly for T2D. ERL for persons with diabetes was computed using mortality rates estimated from models with spline effects of age and calendar time, fitted separately for men and women and by diabetes type. ERL for persons without diabetes was derived from a multistate model with states *no diabetes*, *T1D*, *T2D*, and *dead*, using incidence rates of T1D and T2D and mortality rates for each state. YLL was presented separately for men and women, by diabetes type and age at diagnosis, as a function of attained age (starting at selected ages at diagnosis).

RESULTS

Prevalence

As of January 1, 2025, a total of 366 174 people in Denmark (203 931 men and 162 243 women), or 6.1% of the population, had diabetes (online supplemental tables 1 and 2). Of these, 29 907 (8.2%) had T1D and 336 267 (91.8%) had T2D (including other types of diabetes). The more than threefold increase in individuals living with diabetes since 2000 was driven by an increase in the prevalence of T2D, averaging around 4.2% (95% CI 4.2% to 4.3%) per year for men and 4.1% (95% CI 4.0% to 4.1%) per year for women across all age groups (online supplemental table 5 and online supplemental figure 1). For T1D, overall prevalence decreased by 0.5% (95% CI −0.6% to −0.5%) per year for men and 0.7% (95% CI −0.7% to −0.7%) per year for women, with an increase in people aged under 40 years and a decrease in older age groups (online supplemental figure 1). Figure 1 presents the age-specific prevalence for T1D and T2D in 2005, 2015, and 2025. In 2025, the prevalence of T1D peaked around age 30 years and was relatively stable after this at about 0.7% for men and 0.5% for women; for T2D, the prevalence peaked at around age 80 years at 22% for men and 16% for women.

Incidence

In the period 1996–2024, a total of 567 510 new diabetes cases were recorded, of which 26 210 (4.6%) were classified as T1D and 541 300 as T2D (online supplemental table 3). Overall, the incidence rates of T1D decreased annually by 1.1% (95% CI −1.3% to −0.9%) for men and women during the study period (figure 2, upper right panel, and online supplemental table 5). However, this resulted from a combination of an increase in the younger and a decrease in individuals above 40 years. Moreover, from 2010 onward, the incidence rates of T1D increased for men and women (online supplemental table 5). The incidence rates of T1D peaked at age 14 years for men and 12 years for women (figure 2, upper

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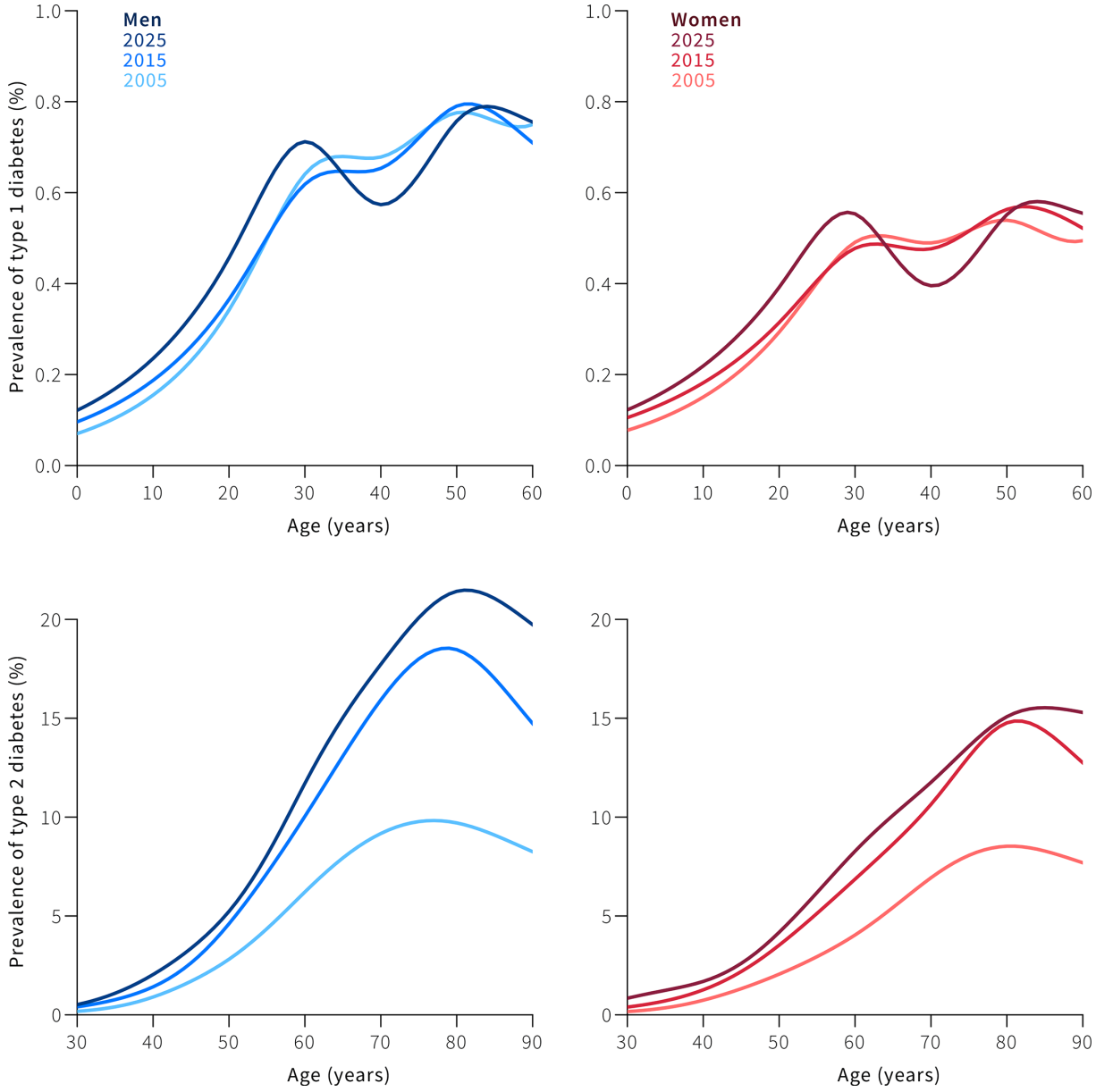


Figure 1 Prevalence of type 1 diabetes (upper panels) and type 2 diabetes (lower panels) every 10 years from 2005 (lighter color) to 2025 (darker color). Men are shown in blue and women in red. Estimates are from separate models for each date, sex, and diabetes type with a natural spline effect of age on the log-prevalence scale. Note the difference in x-axis and y-axis between type 1 diabetes and type 2 diabetes.

left panel). T2D incidence showed a non-linear pattern, declining between approximately 2010 and 2015 and increasing again in recent years, resulting in an overall upward trend from 1996 to 2024 (average increase of 1.3% (95% CI 1.2% to 1.3%) per year for men and 1.1% (95% CI 1.0% to 1.1%) per year for women across the full period). The trend over time was similar across age groups, although the increase in recent years was more pronounced in younger individuals (<50 years), particularly among women (figure 2, lower right panel). Men had higher incidence rates than women across all age groups and calendar years. In 2024, the incidence rates were relatively constant from age 60 years to 80 years at around 10 per 1000 person-years (PY) for men and around 7 per 1000 PY for women (figure 2, lower left panel).

Mortality rates

From 1996 to 2024, a total of 280 327 deaths were recorded among persons with diabetes—24 336 (8.7%) among T1D and 255 991 (91.3%) among T2D (online supplemental table 4). In the same period, there were an estimated 1 318 256 deaths in persons without diabetes. Figure 3, left panel, shows the mortality rates for persons with T1D and T2D by calendar time for selected ages. Overall, mortality rates were higher among men than women. From 1996 to 2024, the mortality rate for T1D increased slightly until around 2008 and hereafter

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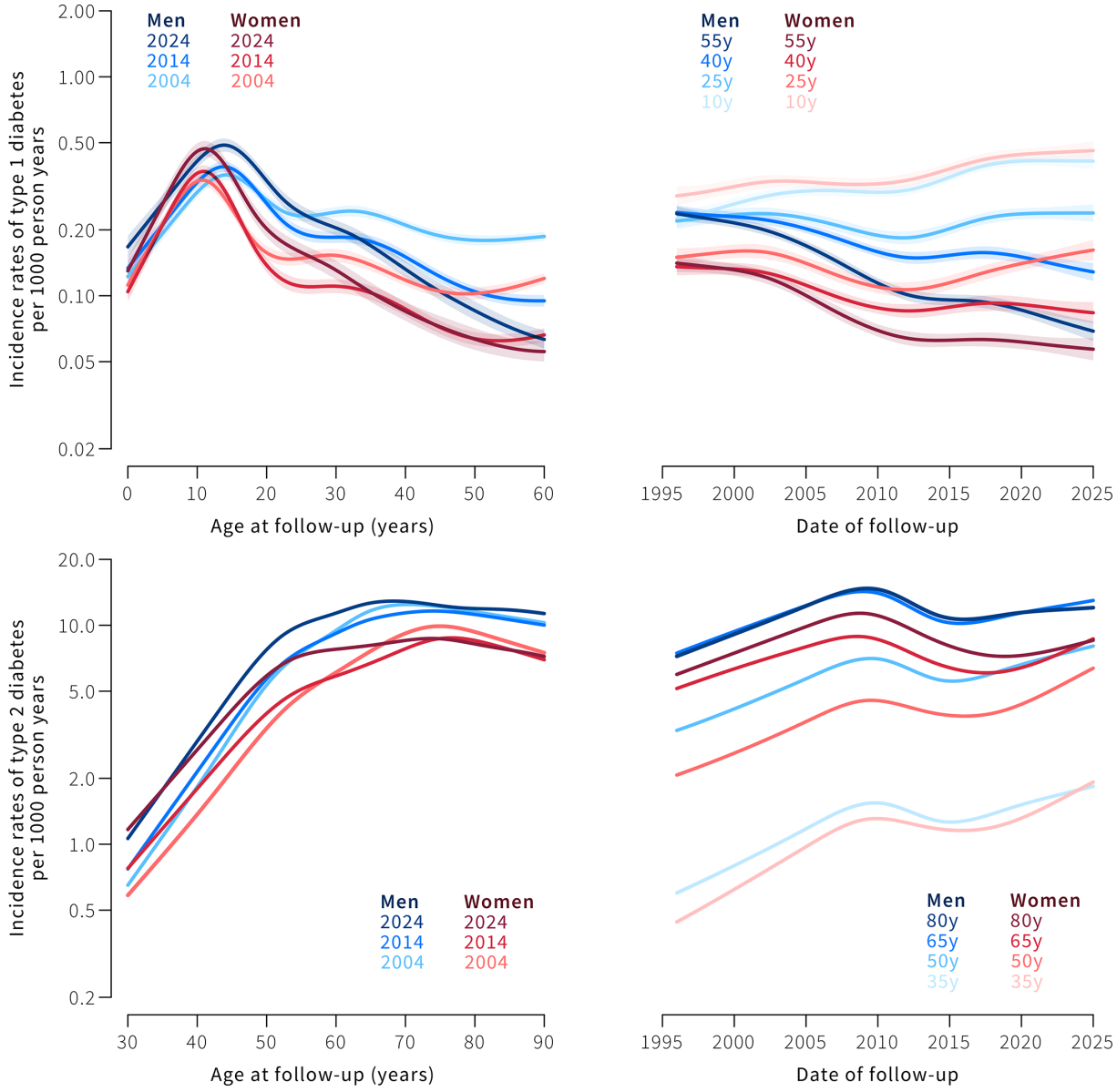


Figure 2 Incidence rates of type 1 diabetes (upper panels) and type 2 diabetes (lower panels) by age for select years (left panels) and by year for select ages (right panel). Men are shown in blue and women in red. Note the difference in x-axis and y-axis between type 1 diabetes and type 2 diabetes. The date of follow-up on the right panels refers to January 1, 1995 to January 1, 2025.

decreased considerably, averaging across the whole period to an annual decrease of 3.0% (95% CI −3.2% to −2.8%) and 2.0% (95% CI −2.2% to −1.7%) per year for men and women, respectively. For T2D, the mortality rate decreased until around 2012 and hereafter remained relatively stable, with an overall annual decrease of 2.4% (95% CI −2.5% to −2.4%) and 1.9% (95% CI −1.9% to −1.8%) per year for men and women, respectively.

Standardized mortality ratio

Figure 3, right panel, shows SMRs by calendar years for T1D and T2D relative to the population without diabetes. In general, SMRs were higher in women than men for both T1D and T2D. Between 1996 and 2008, the SMR for persons with T1D increased, while it declined for persons with T2D. From 2008 to 2024, the pattern reversed, with a decline in SMR for persons with T1D and an increase for persons with T2D. However, for men with T2D, SMR decreased again after 2020. In 2024, the SMRs for T1D and T2D were similar for men and women. SMRs by age from 2008 to 2024 are shown in online supplemental figure 2. For T1D, we observed an increase in SMR from age 30 years to 40 years, and hereafter a steep decrease until age 90 years. In 2024, the mortality rate was markedly higher among persons with diabetes relative to persons without diabetes in younger ages and less pronounced in older ages (online supplemental figure 3).

The annual average change in prevalence, incidence, mortality and SMR for the entire follow-up period and restricted to the period 2010–2024 can be found in online supplemental table 5.

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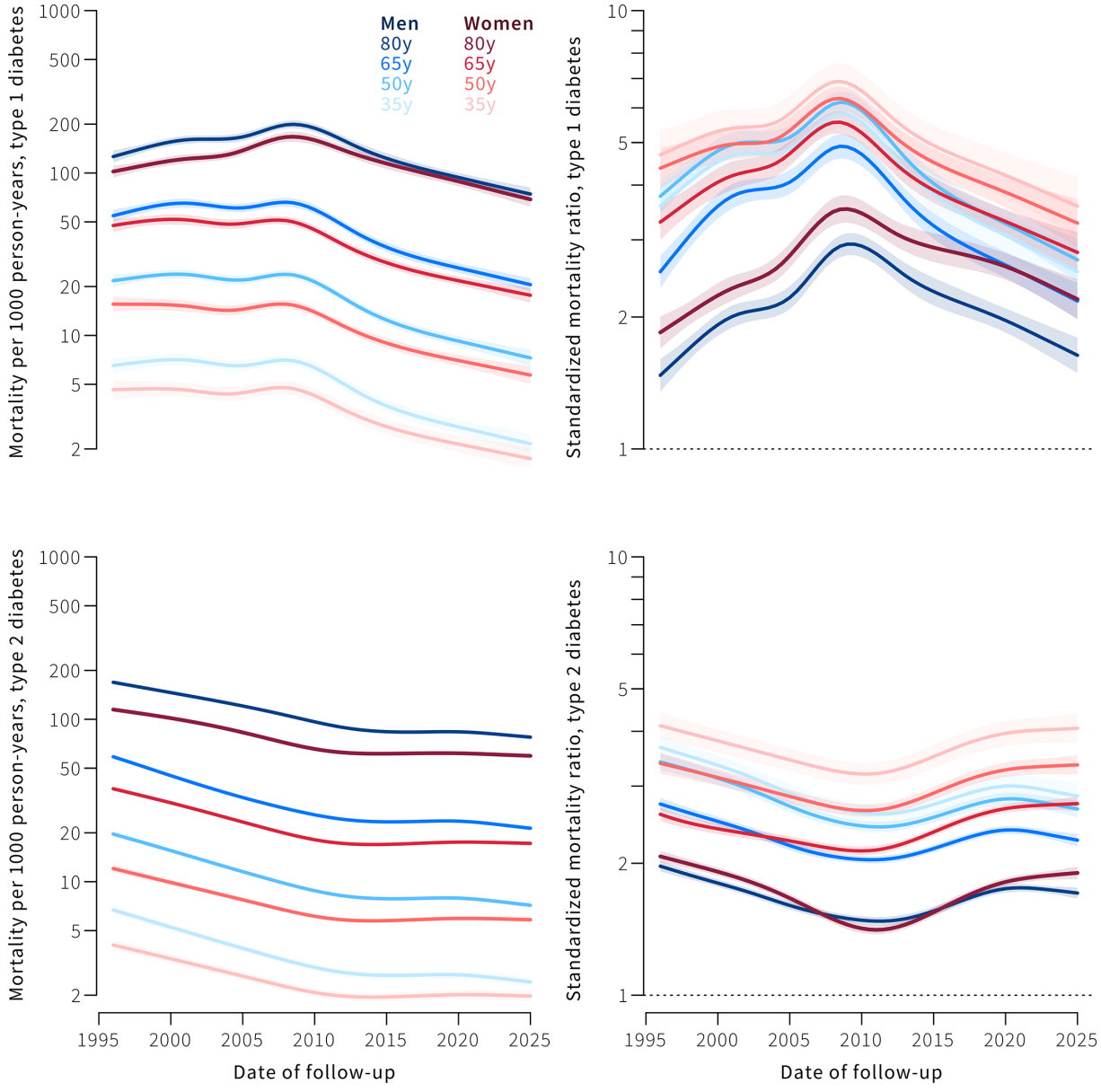


Figure 3 Mortality rates (left panel) and standardized mortality ratios (right panel) of type 1 diabetes (upper panels) and type 2 diabetes (lower panels) by year for select ages. Men are shown in blue and women in red. Note the difference in the y-axis between the left (mortality rates) and right (standardized mortality ratios) panels. The date of follow-up refers to January 1, 1995 to January 1, 2025.

T1D versus T2D mortality rate ratio

We observed a higher mortality rate among persons with T1D compared with those with T2D, with an increase in mortality rate ratio until ages 55–60 years, and then a decrease for men, while women with T1D remained higher than T2D (online supplemental figure 4). On average, the mortality rate ratio increased until 2009 at a level of around 2 (two times higher mortality rate in persons with T1D compared with T2D) but subsequently amounted to a similar mortality rate ratio between T1D and T2D in 2024.

Lifetime risk and lifetime lost to diabetes

Lifetime risk, expected lifetime spent with diabetes, and lifetime lost to diabetes are shown in online supplemental table 6. In 2025, the lifetime risk of T1D was 1.4% for men and 1.1% for women, while the lifetime risk of T2D was 37% for men and 31% for women.

For both T1D and T2D, the YLL to diabetes decreased with increasing age at diagnosis and was markedly higher in women with T1D compared with T2D (online supplemental figure 5). However, with increasing calendar year, the difference between T1D and T2D decreased. In 2024, the difference was similar in men, but in women, the YLL was higher for T1D compared with T2D. The figure also shows that the effect of age at diagnosis is small, as the curves closely follow each other regardless of their starting point. Online supplemental figure 6 shows the YLL curves for individuals diagnosed at age 50 years

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across different calendar years. For both men and women with T1D, the YLL increased from 2000 to 2010, followed by a substantial decline from 2010 to 2025, indicating fewer expected YLLs at age 50 and beyond. In contrast, for T2D, YLL slightly decreased from 2000 to 2010, then increased, with curves converging and remaining similar between 2015 and 2025. In 2025, YLL to T1D at age 50 was 4.9 years for men and 6.9 years for women. For T2D, YLL at the same age was 4.9 years for men and 5.7 years for women (online supplemental table 6).

DISCUSSION

In this nationwide study based on Danish health registers, we provide up-to-date estimates on prevalence, incidence, mortality, and lifetime lost associated with T1D and T2D over three decades. We found an increase in T1D prevalence and incidence among younger age groups (<40 years) and a decrease in older age groups, whereas T2D prevalence and incidence increased over time, with larger increases in prevalence and incidence among younger age groups. After 2008, the mortality rates decreased for T1D but stabilized for T2D after 2012, with consistently higher rates in men. While excess mortality relative to individuals without diabetes declined for T1D after 2008, it increased slightly for T2D after 2010. Lastly, the amount of lifetime lost to T1D was reduced in the period after 2010 and remained stable for T2D.

The increasing incidence and prevalence of T1D in the younger age groups are consistent with trends reported internationally.[22] The observed decline in incidence and prevalence in older ages may reflect a misclassification of T2D as T1D, especially in the earlier years of the diabetes register, due to changes in diagnostic criteria with a higher tendency to diagnose insulin-treated persons with T2D as T1D in the past. Recent global estimates show both increasing and decreasing incidence of T1D in older adults, underscoring the diagnostic uncertainty in this age group.[23 24] With the increasing incidence of T1D among youth, Denmark continues to rank among the countries with the highest reported global rates, alongside other high-income countries, particularly the other Nordic countries,[22 25] possibly also reflecting a higher level of case registration.

We observed an average overall increase in incidence and prevalence of T2D. After a decline in the incidence from around 2010 to 2015, the incidence rates of T2D resumed an upward trend in recent years. The decline observed after around 2010, which has also been observed in other high-income countries,[4] may partly reflect the shift from oral glucose tolerance test (OGTT) and fasting plasma glucose (FPG) test-based testing to HbA1c in 2011. Individuals who have previously been diagnosed with T2D using OGTT or FPG test may not have met the HbA1c threshold, leading to missed or delayed diagnoses.[26] However, the decreasing trend has not been consistently observed in all countries that adopted HbA1c testing.[4] More recently, some countries have reported a renewed increase in T2D incidence, particularly among younger age groups,[25 27 28] although newer data remain scarce. The current upward trend may represent a compensatory increase following the shift to HbA1c, but also a genuine increase in occurrence. Multiple factors may contribute, including lifestyle and environmental changes such as increased sedentary behavior, reduced physical activity, and higher consumption of energy-dense, ultra-processed foods. Together, these factors contribute to rising obesity rates,[29] the major driver of T2D.[30] Conversely, the convenience of HbA1c testing likely facilitated broader screening. Additionally, the establishment of the Steno Diabetes Centers across Denmark may have contributed to greater awareness and screening. As of January 1, 2025, the overall prevalence of T2D in Denmark was 6.3% for men and 5.0% for women, consistent with prevalence data reported from other European countries in recent years and remain among some of the lowest reported in the world.[28 31]

All-cause mortality among individuals with T1D has continued to decline. This trend aligns with patterns observed in other countries, including Scotland, Spain, Australia, Latvia, USA (Kaiser Permanente Northwest), Sweden and Finland.[32–35] Importantly, the relative mortality rate compared with persons without diabetes has also decreased considerably. In 2025, lifetime lost to T1D was 4.9 years for men and 6.9 years for women aged 50 in Denmark. Differences in methodology and reporting make comparisons between studies difficult, and more recent data in T1D are lacking. However, data from Finland found that by 2017, the life expectancy at age 20 for individuals with T1D was 9.9 years lower than for the general population,[35] in range with our numbers in these years. A study from Sweden among adult-onset T1D in 2014 found up to 5 YLL depending on age at onset, but found no overall significant lifetime lost to T1D; however, this was a small study conducted in one county of Sweden.[32] The decline in mortality may reflect improvements in diabetes management and care over the past decades, like advances in insulin analogs, insulin pumps, and continuous glucose monitoring, as well as treatment of cardiovascular and kidney risk factors (blood pressure and lipids) and a decline in smoking.[9 10] Despite these gains, persons with T1D still face a two to six times higher mortality rate in 2024 before age 70 compared with those without diabetes. This excess mortality is particularly pronounced among women, highlighting the need for continued efforts to improve long-term outcomes. The development of treatments reducing cardiovascular and kidney outcomes in T2D has not yet been followed by similar studies in T1D,[36] but studies are ongoing.[37]

Among individuals with T2D, mortality declined from 1996 to around 2012 and thereafter remained relatively stable until 2024. Similar patterns have been observed in other high-income countries, where mortality declines have slowed or plateaued in recent years, although the exact timing of these changes varies between studies.[38 39] In the population without diabetes, the mortality has also

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been declining, and we observed that mortality relative to the population without diabetes increased slightly since 2010. The COVID-19 pandemic may also have influenced the SMR patterns in the most recent years of follow-up, although the excess mortality was relatively low in Denmark.[40] Lifetime lost to T2D was estimated to be 4.9 years for men and 5.7 years for women aged 50 years in 2025 with very little effect of age at diagnosis, broadly comparable to that observed in other countries. A recent study from the Emerging Risk Factors Collaboration combining data from 19 high-income countries up to 2019 estimated that a person with T2D aged 50 lost 6–14 years of life depending on age at diagnosis, with about 3–4 years of reduced life expectancy per decade of diabetes.[41] Overall, we found higher prevalence, incidence and mortality rates among men than women for both T1D and T2D, consistent with findings from the Global Burden of Diabetes study. Nevertheless, women appeared to experience a greater relative burden of diabetes, reflected by more life years lost and greater excess mortality relative to women without diabetes, despite lower absolute mortality rates than men.

The lifetime lost to both T1D and T2D remains clinically significant despite broader use of cardioprotective medication, enhanced screening, and management of complications of diabetes in Denmark over the past decade. One possible explanation may be health improvement in the general population, occurring more rapidly compared with those with diabetes, thus widening the mortality gap and hence increasing the lifetime lost.[42] Increased use of cardioprotective treatment such as statins and antihypertensive medicine, as well as declining smoking prevalences, may have contributed to lower mortality both among those with and without diabetes.[43] In addition, the cardiovascular and renal protective effects of SGLT-2 inhibitors and GLP-1 receptor agonists were only demonstrated after 2015 and were incorporated into international and national treatment guidelines from 2018 onward. Implementation, however, has been incomplete. Several studies have shown that uptake was initially slow, and that cardiovascular risk status was not consistently associated with receipt of guideline-directed, organ-protective therapy.[44 45]

Strengths and limitations

A major strength of our study is its nationwide coverage, encompassing the entire Danish population, enabling analyses with 29 years of follow-up. We used reliable data from comprehensive Danish registries, known for their validity and completeness. An important strength of the algorithm defining time and type of diabetes is the inclusion of clinically validated information from DADD, as well as diabetes-specific medication and data on podiatry and laboratory measurements. This allowed us to identify individuals with T2D, including those not receiving pharmacological treatment. The use of multiple data sources to identify individuals with diabetes increased the sensitivity, which was particularly important for capturing persons with T2D treated in general practice but without a diabetes diagnosis in NPR. Compared with the previous version of the algorithm,[6] the addition of HbA1c measurements has led to a growing proportion of individuals being identified based on HbA1c criteria over time. The risk of false positives is considered low, as all data sources contain diabetes-specific information. Our data also allowed valid differentiation between diabetes types, enabling separate analyses for both T1D and T2D, especially after 2005 with the inclusion of DADD data, which is considered more accurate than only relying on NPR diagnoses. An algorithm similar to ours, but without inclusion of DADD data, demonstrated high overall performance in a recent validation study against survey data,[46] suggesting that our algorithm may also achieve high performance.

However, the study has some limitations. Classification of diabetes type based solely on register data is inherently imprecise. To mitigate this, we applied multiple criteria rather than relying exclusively on International Classification of Diseases (ICD) codes or insulin redemptions.[47] The algorithm is designed to identify T1D with high specificity, meaning those not meeting the criteria for T1D are classified as T2D. In Denmark, most individuals with T1D are managed in outpatient clinics and therefore have a corresponding ICD diagnosis code, although some are treated exclusively in primary care and must be identified through alternative criteria. Among individuals younger than 30 years, insulin treatment alone was considered sufficient for T1D classification to maintain high sensitivity, as requiring oral glucose-lowering medication such as metformin could increase the risk of misclassifying younger individuals with T2D. Nevertheless, some degree of misclassification may still have occurred, particularly in the years before 2005 and among individuals who died before DADD was established, potentially overestimating T1D cases in older age groups in the first period. We excluded purchases of metformin only in women between ages 20 and 40 on suspicion that these were related to PCOS rather than diabetes. This may have the effect that a number of women are included in the register shortly after turning 40, thereby artificially increasing the incidence for women in that age category. Misclassification may also exist for people with subtypes. For instance, individuals with latent autoimmune diabetes in adults, which lacks a specific ICD-10 code, may initially be diagnosed as T2D. Similarly, other diabetes subtypes (ICD-10: E12–E14) were grouped with T2D in the analyses. Additionally, the completeness of the underlying data sources has varied over time. National laboratory data were available from 2011 but considered nationwide complete only from 2015 onward, and reporting from general practitioners to DADD decreased between 2014 and 2017, which may have affected the number of cases in these years. From 2022, DADD was converted to an algorithm-based database, like our algorithm, not relying on clinical validation, thereby losing its previous classification advantages. Also, individuals registered with diabetes before

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January 1, 1996 were excluded from the duration-based analyses because the date of diabetes onset before this period could not be reliably determined from the available register data. Consequently, the duration-based analyses were effectively left truncated in the earliest study years, potentially leading to underestimation of mortality associated with longer diabetes duration. Lastly, our estimates reflect diabetes identified through register data and therefore do not include undiagnosed cases in the general population. Consequently, the true prevalence and incidence of diabetes are likely to be higher.

CONCLUSIONS

Prevalence and incidence of T1D continue to rise moderately at younger ages, while T2D prevalence and incidence show an overall increase over time despite a temporary decline in the early to mid-2010s. Lifetime risk remains low for T1D but high for T2D, and although mortality among people with diabetes has declined over time, it remains elevated compared with the general population. Life expectancy is still reduced, with an average loss of 5–7 years depending on sex and diabetes type. Overall, these trends indicate improved survival and health among individuals with diabetes both absolutely and relative to the population without diabetes, yet a substantial gap persists compared with those without the disease.

Contributors RW, VK, SB, CL, PR, BC, and PFR designed the research. BC performed the statistical analysis. RW and PFR wrote the paper. RW, VK, SB, CL, PR, BC, and PFR edited the paper. BC, RW, and PFR had the primary responsibility for the final content. All authors reviewed and approved the final manuscript. BC, RW, and PFR are the guarantors of this study and, as such, had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analyses.

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Patient consent for publication Not applicable.

Ethics approval This study involves human participants, and approval for data access and usage of personal data was obtained from The Danish Health Data Authority (reference numbers FSEID-00007492 and FSEID-00007493). According to Danish law, ethics approval and participant consent are not required for registry-based studies in Denmark.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data may be obtained from a third party and are not publicly available. The data for this study are population-wide registers, placed at our disposal on the servers of Statistics Denmark. They are barred from release to the public on grounds of confidentiality. Restrictions apply to the accessibility of these data, which were analyzed under license for this study.

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